SIDDHANT SINGH

Bengaluru, India | +91-8747893867 | siddhant3103@gmail.com
linkedin.com/in/siddhant-singh | github.com/SID-singh1 | leetcode.com/u/_SID_SINGH/

EDUCATION

Vellore Institute of Technology (VIT), Vellore | B.Tech, Computer Science & Engineering | CGPA 8.99 *2022 – 2026*

EXPERIENCE

AI/ML Research Intern, Samsung R&D Institute India *Jan 2026 – Aug 2026*

- Built an on-device document classifier for budget Android phones: EfficientNet-B4 teacher distilled into a MobileNetV2 student, with INT8 TFLite quantization. The final model is **2.87 MB with 97.87% clean-test accuracy** (teacher 95.09%) and ~4.35 ms inference latency.
- Engineered an offline Android app supporting gallery and live-camera inputs with automated folder classification; leveraged hardware acceleration and multithreading to achieve **<10 ms** latency on NPU devices.
- Measured real-world accuracy on 486 phone-captured images (**84.1%** confidence-filtered vs 97.9% clean); raised worst-class recall from **8.3% to 69.4%** through targeted data collection.

PROJECTS

Distqueue: Distributed Task Queue | Python, Redis Streams, Lua, Docker, Prometheus

- Built a job queue from scratch on Redis Streams: producers enqueue to a stream, consumer-group workers process and ack, and separate scheduler (delayed jobs) and monitor (dead-worker recovery) services handle the rest. At-least-once delivery, atomic MULTI/EXEC enqueue; **~6,200 enqueues/s (p99 0.37 ms)** and **~2,000 jobs/s** per worker on local Redis.
- Detected dead workers via daemon-thread heartbeats (5 s refresh, 15 s TTL) and reclaimed jobs with XAUTOCLAIM. A chaos test SIGKILLs a worker mid-job and a survivor completes it.
- Prevented delayed-job double-injection with an atomic Lua script. Retries use exponential backoff with jitter and a dead-letter queue.
- Wrote **52 tests** (unit, integration, chaos) at **~95% core coverage**. Added Prometheus + Grafana via Docker Compose.

Prompt Memory: RAG-Powered Prompt Engine | FastAPI, Qdrant, MongoDB, SSE, Chrome MV3, GitHub Actions

- Built a Manifest V3 Chrome extension and FastAPI backend that rewrite prompts in ChatGPT, Claude, Gemini and Perplexity using per-user vector memory (Qdrant + MiniLM), streamed over SSE.
- Implemented a Google OAuth2 popup flow with token polling and JWT auth, per-tier daily quotas, and a custom sliding-window rate limiter keyed on user ID (IP-based limits failed behind the Hugging Face Spaces proxy).
- Designed LLM failover (Groq key rotation on 429, then Gemini, then OpenRouter) with automatic eviction of decommissioned models, after an upstream deprecation caused a silent outage.
- Enforced tenant isolation with a mandatory user_id filter on every vector query: **0 cross-user leaks** and **91.7% recall@k** on a 24-case eval.
- Shipped an HMAC-secured private analytics dashboard (anonymized user IDs), structured JSON logging, and health endpoints. Wrote **443 tests (61% backend coverage)** with GitHub Actions CI that auto-deploys to Hugging Face Spaces.

Vaani: Hindi-Hinglish Voice-Notes Service | FastAPI, PostgreSQL, JavaScript, ffmpeg, llama.cpp

- Built a full-stack voice-notes service: browser or Telegram audio goes through ffmpeg (webm to 16 kHz wav) and Whisper transcription, then an LLM returns a summary, action items and sentiment as strict JSON, stored in PostgreSQL.
- Developed an async FastAPI REST backend with IP rate limiting, a health endpoint, and login-free per-user history (UUID provisioning). Added a key-protected admin dashboard (total users, requests, 7/14-day activity) and a vanilla-JS frontend with Web Audio waveform.
- Fixed out-of-memory errors on audio over 30 s via sliding-window chunking. Deployed a Telegram bot on Render, with a config switch between on-device inference (INT8/INT4: footprint **down 68.7%**, generation **+211%**) and cloud APIs.

SKILLS

Languages: Python, C, C++, Java, SQL, JavaScript

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS (SQL & NoSQL), Operating Systems, Distributed Systems, Concurrency

Backend & Databases: FastAPI, Spring Boot, REST APIs, SSE, OAuth2/JWT, Redis (Streams, Lua), MongoDB, PostgreSQL, Qdrant, Chrome Extensions (MV3), pytest

Cloud & DevOps: AWS, Linux (CLI), Git, GitHub, Docker, Docker Compose, GitHub Actions (CI/CD), Prometheus, Grafana, Hugging Face, Render

AI/ML: LLMs, Transformers, RAG, prompt engineering, agentic AI (LangChain, LangGraph), knowledge distillation, quantization, TFLite, ONNX Runtime

ACHIEVEMENTS & CERTIFICATIONS

- LeetCode: **700+ problems solved, contest rating 1837** | VIT Top Coders: **top 200 out of 15,000**
- **AWS Certified Cloud Practitioner**